

United States Artificial Intelligence (AI) Action Plan

By Irving Martinez Pacheco, Ph.D.

1. Introduction and Objectives

Context and Purpose

On January 23, 2025, the President's executive order determined the need for an artificial intelligence (AI) action plan to ensure the leadership of the United States in this area. This plan describes priority activities to improve innovation, to drive economic growth, to ensure security and ethics in AI applications and to promote business under the leadership of AI. Using Generative AI, using advanced fast techniques and interference with the integration of the curriculum without changing its structure, the US can accelerate the development of newly driven startups. These start -ups will not only create local employment and economic opportunities but will also nominate the country as a world leader in Sustainability solutions. In addition, this plan emphasizes how adaptation to Sustainable Development Goals (LDG) can improve US leadership AI and easy, expanding AI -based startups to international markets that are increasingly in need of sustainability. Generative AI's strategic use will serve as a catalyst to achieve these goals in education and business, giving students and educators the opportunity to develop scalable, sustainable solutions that improve US economic forces and competitiveness in global markets.

Main Objectives

1. Maintain US leadership in AI, promoting policies that promote innovation, economic expansion and global competitiveness through generative AI applications.
2. Remove unnecessary regulatory barriers that may limit the development of AI, providing carefree generative Generative AI solutions throughout education and entrepreneurship.
3. Promote generative AI applications in education and entrepreneurship to provide students, educators and institutions with tools to develop sustainable companies with great influence that meets the requirements of the international market.
4. Create a powerful AI -led start -up ecosystem in educational institutions to create employment, improve economic expansion and develop SDG -adjusted solutions driven by generative AI.
5. Promote the performance of the international market for AI by supporting new businesses in line with sustainability and AI observation standards needed for the global economy, ensuring that US entrepreneurs are located as leaders in AI -led sustainability solutions.

2. AI Application and Use

Key Sectors

Private sector: Promote widespread use of Generative AI in education and business to manage technological innovations, strengthen business competence and optimize institutional efficiency. By integrating AI drive automation and predictable analysis, private organizations can improve operational capabilities and develop AI -managed business models that meet sustainability requirements.

Public Sector: Optimization of Generative AI in government -backed education and business programs to improve educational methods, introduce political frameworks and support the establishment of newly managed new businesses in academic institutions. By integrating the AI public sector initiatives, decision-makers can speed up the adoption of sustainable business solutions for AI.

Higher education and entrepreneurship: The introduction of Generative AI in higher education institutions to promote AI -leading innovation and establishing globally -scalable start -ups, focusing on sustainability and market expansion. Universities and colleges will integrate generative AI tools in training programs, business programs and research initiatives to develop future business executives with AI competence. The AI-Improved Learning Environment will provide students with the technical and business skills needed to establish, manage and scale companies that meet the LDG and international legislation.

3. Explainability and Assurance of AI Model Outputs

To ensure the responsible and trustworthy deployment of generative AI, the following measures will be implemented:

- **Standardized Guidelines for AI Transparency:** Develop and enforce regulatory standards that require AI-driven applications in education and entrepreneurship to disclose how decisions are made, ensuring that outputs are interpretable and traceable.
- **Independent Audits and Compliance Mechanisms:** Establish third-party review processes for AI models, particularly those used in high-stakes applications such as education and sustainability. This will include bias assessment, security validation, and compliance verification with ethical AI frameworks.
- **Real-World AI Testing and Certification:** Create AI certification programs that evaluate model accuracy, reliability, and ethical adherence before deployment. These programs will help educational institutions and businesses confidently integrate AI solutions that align with global sustainability and compliance standards.
- **User-Centric AI Assurance:** Implement AI explainability tools that allow educators, students, and entrepreneurs to understand and interpret AI-generated recommendations. Interactive dashboards and AI transparency reports will be developed to increase confidence in AI-driven decision-making.

- **AI Risk Management Strategies:** Introduce proactive risk assessment methodologies that identify and mitigate potential biases, unintended consequences, and ethical concerns in AI applications. AI-powered entrepreneurship initiatives will be required to demonstrate responsible AI governance as part of their business development strategies.

4. Education, Workforce, and AI Entrepreneurship

Academic Programs: Building the Next Generation of AI Entrepreneurs

To accelerate the adoption of AI in entrepreneurship and sustainability, higher education institutions must expand initiatives that provide hands-on training in generative AI applications, AI-driven business models, and sustainable innovation.

1. Interdisciplinary AI Programs: Institutions of higher learning ought to provide interdisciplinary AI programs that combine technology, sustainability, and business. These courses will give students the know-how to use AI to innovate, allowing them to create AI-powered startups or use AI to improve already-existing businesses.

2. Industry Partnerships: To develop internship and apprenticeship programs, partnerships between private businesses and higher education institutions should be encouraged. Students will be able to match their studies with industry demands by taking advantage of these possibilities to obtain practical experience in AI-driven business settings.

3. Using AI to Promote Sustainability Education: To ensure that graduates understand how to apply AI for environmental, social, and economic benefit, curriculum design should incorporate courses that concentrate on AI applications in sustainability.

AI Sustainability Toolkit for Education: Empowering AI-Powered Entrepreneurship

A comprehensive AI Sustainability Toolkit should be developed to provide educators and students with structured resources to integrate generative AI into their learning environments.

Higher Education Startups: Establishing AI Innovation Hubs and Incubators

To foster a culture of AI-driven innovation within higher education, institutions should establish AI innovation hubs and startup incubators.

1. AI Startup Accelerators: These accelerators will offer student-led AI businesses financial incentives, technical assistance, and business mentoring. Programs must be created to assist new businesses at every step of growth, from conception to expansion.

2. Access to Funding and Mentorship: AI incubators ought to provide students with AI-driven company growth techniques, funding opportunities, and mentorship from AI specialists.

3. Networking and Policy Engagement: To link AI entrepreneurs with financiers, legislators, and sustainability specialists, specialized networking initiatives have to be established. This will give student-led entrepreneurs the assistance they need to expand and build their AI-powered companies.

Building a Future-Ready Workforce

By integrating AI education, industry collaborations, and entrepreneurship training, the U.S. can develop a robust pipeline of AI-driven entrepreneurs. This ecosystem will equip students and professionals with the skills and knowledge to address sustainability challenges using AI-powered solutions, positioning the U.S. as a global leader in AI-driven sustainable innovation.

Expanding AI-Driven Opportunities

To further strengthen AI-driven entrepreneurship and workforce readiness, higher education institutions should:

1. **Create AI-Specific certificates:** To give professionals and students credentials accepted by the industry, offer specific certificates in sustainability and AI entrepreneurship.
2. **Strengthen AI Research & Development:** Fund R&D initiatives that investigate novel AI uses in corporate innovation and sustainability.
3. **Encourage AI-Driven Public-Private Partnerships:** To advance AI-powered economic growth and sustainability projects, encourage cooperation between governmental organizations, businesses, and universities.

5. Innovation and Competition

Removing Barriers to AI-Driven Business Creation

To foster a dynamic and competitive AI-driven economy, regulatory frameworks should be adapted to facilitate innovation while ensuring ethical AI use.

1. **Regulatory Modernization:** Eliminate onerous regulatory obstacles that impede the development of AI-driven businesses and guarantee that generative AI solutions may be applied with little difficulty in the entrepreneurship and higher education sectors.
2. **Ethical AI Standards:** Provide precise moral principles to control AI creation and use while preserving creative freedom.

Incentivizing AI-Driven Entrepreneurship

To encourage the development and expansion of AI-driven startups, targeted incentives should be introduced.

1. **Tax Incentives and Grants:** Provide tax exemptions, grants, and low-interest loans to help firms that use generative AI to enter foreign markets, boost the economy, and have a lasting impact.
2. **Commercialization & Research in AI Grants:** Ensure that creative solutions get from academia to the industry by providing funds for AI research with a clear route to commercialization.

Strengthening AI Innovation Ecosystems

Collaboration between higher education institutions, research centers, and private-sector investors should be reinforced to accelerate AI innovation.

6. Intellectual Property and AI Innovation Protection

Removing Barriers to AI-Driven Business Creation

To foster a dynamic and competitive AI-driven economy, regulatory frameworks should be adapted to facilitate innovation while ensuring ethical AI use.

1. **Regulatory Modernization:** Remove restrictive regulatory barriers that hinder AI-driven business creation and ensure generative AI solutions can be implemented across higher education and entrepreneurship sectors with minimal friction.
2. **Ethical AI Standards:** Establish clear ethical guidelines to govern AI development and deployment while maintaining flexibility for innovation.

Incentivizing AI-Driven Entrepreneurship

To encourage the development and expansion of AI-driven startups, targeted incentives should be introduced.

1. **Tax Incentives and Grants:** Implement tax breaks, grants, and low-interest loans to support startups utilizing generative AI for sustainable impact, economic growth, and international market entry.
2. **AI Research & Commercialization Grants:** Provide funding for AI research with a direct path to commercialization, ensuring innovative solutions transition from academia to the market.

Strengthening AI Innovation Ecosystems

Collaboration between higher education institutions, research centers, and private-sector investors should be reinforced to accelerate AI innovation.

1. **AI Innovation Hubs:** To promote technical advancements and commercialization, set up specialized AI innovation hubs that link investors, entrepreneurs, and researchers.
2. **Public-Private Partnerships:** To improve funding for AI research, workforce development, and company incubation initiatives, academia, government, and industry should fortify their partnerships.
3. **Global AI Competitiveness:** Create regulations that make the United States a pioneer in AI-powered entrepreneurship and guarantee that homegrown businesses can compete in global marketplaces.

AI Intellectual Property and Legal Frameworks

To support AI entrepreneurs, intellectual property (IP) laws must be modernized to reflect the evolving nature of AI innovation.

1. **AI-Specific IP Laws:** Update intellectual property regulations to guarantee that inventions, business models, and content produced by AI are properly protected.
2. **AI Patent Frameworks:** Create patent frameworks tailored to AI that minimize needless legal complications while granting entrepreneurs ownership rights over AI-driven procedures, algorithms, and goods.
3. **Legal Consultation for AI Entrepreneurs:** Create programs for legal consultation that give AI entrepreneurs advice on worldwide and U.S. intellectual property laws, especially those pertaining to sustainability and AI-driven industries.
4. **International AI IP Cooperation:** To guarantee that American AI firms can grow internationally while preserving their discoveries, promote international cooperation on AI-related intellectual property laws.

7. Public Procurement and AI Acquisition

- **Modernize Public Procurement Frameworks:** Revise government procurement regulations to provide AI-powered solutions that improve entrepreneurship and sustainability in education top priority. Provide precise rules for assessing and implementing generative AI in public sector projects.
- **AI-Powered Grants & Incentives:** Establish tax breaks and financing initiatives for start-ups and SMEs creating AI-powered learning resources with quantifiable sustainability results. Promote government-supported centers for AI innovation.
- **Public-Private AI Integration:** To test and expand AI-powered business models in public education and entrepreneurial ecosystems, form strategic alliances

between governmental organizations, academic institutions, and private AI developers.

- **Regulatory Sandboxes for AI Procurement:** Put in place programs that enable controlled testing of AI procurement models in company development and education.

8. International Collaboration

- **Global AI Ethics Leadership:** Lead efforts for AI governance that prioritize sustainability, equity, and transparency, bringing U.S.-led frameworks into compliance with global sustainability and AI ethics norms.
- **AI Market Expansion Strategies:** Offer financial assistance and regulatory advice to American AI startups looking to expand internationally with SDG-compliant AI solutions.
- **Cross-Border R&D Centers for AI:** Through bilateral and multilateral agreements, promote international research collaboration, especially in the areas of AI-driven entrepreneurship and sustainable education.
- **AI for Sustainable Trade:** Promote AI-driven sustainability criteria in international trade agreements to make sure AI solutions support the circular economy, emissions reduction, and environmentally friendly corporate practices.

9. Economic Impact and Global Growth

- **AI-Driven Economic Metrics:** Create KPIs tailored to AI to track the growth of AI-powered companies, productivity gains, and job creation. Collaborate with global organizations to standardize evaluations of the economic impact of AI.
- **Sustainable AI Investment Policies:** By offering tax exemptions, government matching programs, and green funding channels, financial institutions and venture capitalists are encouraged to invest in AI-driven sustainability firms.
- **AI-Powered Workforce Development:** Create national upskilling initiatives to guarantee that workers are prepared for employment involving AI, lowering the risk of displacement and fostering economic mobility in AI-driven sectors.
- **AI in Global Trade and Competitiveness:** Promote the inclusion of AI-driven sustainability projects in free trade agreements, thereby enhancing American leadership in AI-driven Net Zero and circular economy models.

10. Next Steps

1. **Pilot Programs:** To confirm the impact of generative AI on innovation and the economy, start focused AI-driven entrepreneurship pilot initiatives in academic institutions.

2. Impact Assessment and Policy Adjustments: To maximize AI-enabled economic advantages, optimize tactics, and modify policies, conduct thorough assessments of generative AI-driven initiatives.

3. Monitoring and Compliance Framework: Provide a flexible framework for monitoring advancements, guaranteeing that AI-powered postsecondary education and entrepreneurial initiatives stay in line with changing market conditions and sustainability norms.